SIMATS ENGINEERING
ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1521 .

COURSE OUTCOME COVERED
CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**
Question Count: 5

Code of Conduct
I harendra (192521215) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: ____________

Assessment Tasks
Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud.
   The system must handle millions of concurrent users with minimal buffering.
   Design a cloud architecture using scalability and caching techniques.
   Explain how load balancing and auto-scaling can ensure smooth streaming.
   Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers.
   They face challenges in maintaining consistency and managing deployments.
   Propose a solution using container orchestration and multi-cloud strategies.
   Explain how service discovery and scaling can be handled efficiently.
   Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
   The organization needs a robust backup and disaster recovery strategy.
   Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

1) Media company - video streaming platform:

Cloud Architecture :-

→ use cloud servers to host videos.

→ store videos in cloud storage.

→ use CDN (Content Delivery Network) to deliver videos faster.

→ use cache to reduce buffering.

Load Balancing and Auto-Scaling:-

→ load balancer sends users to different servers.

→ Auto scaling adds more servers when traffic increases.

→ Removes extra servers when traffic decreases.

Code optimization :-

4! → use pay-as-you-go pricing.

→ store old videos in low-cost storage.

2) Fintech startup - Multi cloud containers:

soln :-

→ use docker containers.

→ use kubernetes for containers orchestration.

→ Deploy apps on multiple cloud providers.

Service Discovery and scaling:-

→ Kubernetes automatically find services.

→ it increases or decreases containers based on demand.

Benefits:-

→ High availability.

→ Better reliability.

→ Avoids vendor lock-in.

Risks:-

→ More complex management.

→ Higher network cost.

→ security management difficult.

3) E-learning platform - Backup and Disaster Recovery.

soln :-

→ Taken regular cloud backups.

→ Enable data replication.

→ use versioning to recover deleted file.

RTO and RPO :-

→ RTO :- Restore the system quickly.

→ RPO :- Restore the latest available data with minimum loss.

Business Continuity steps :-

→ keeps backups in different region.

→ Test Recovery regularly.

→ use automatic failover.

4) Government Agency - Hybrid cloud.

Hybrid Architecture :-

→ keep sensitive data on -premises.

→ store non-sensitive data in the public cloud.

→ connect both using secure VPN.

Identity management and Encryption:-

→ use Multi-factor Authentication (MFA).

→ Encrypt data during storage and transfer.

→ Gives users role-based access.

Challenges:-

→ Complex management.

→ security and compliance issues.

→ Network latency.

5) SaaS provider - High Availability.

→ Deploy the apps in multiple cloud regions.

→ If one region fails, another region continues the service.

Data Replication and failover:-

→ copy data to all regions.

→ use automatic failover to switch users to a healthy region

Monitoring and Alerting:-

→ use cloud monitoring tools to detect issues early.

→ Monitor server health continuously.